CSA0711 – COMPUTER NETWORKS

Common Assignment Report

Analyze Protocol Behaviour and Measure Network Performance Across Alternative Topologies Using Simulation

Course Outcomes Mapped: CO3, CO4, CO5

Bloom's Taxonomy Level: L5 – Evaluate; L6 – Create

SDG Mapping: SDG 9 – Industry, Innovation and Infrastructure

Team Members:

1. Thanigaivel S(Reg. No: 192565006)
2. Rathnakr P (Reg. No: 192425365)

Slot: D   Date: 01.09.2026

1. Problem Statement and Problem Formulation

Modern campus and enterprise networks carry heterogeneous application traffic — web browsing, file transfer, DNS lookups, real-time and bulk data transfer — over infrastructures that combine multiple LAN segments, switches, routers, and end devices. The behaviour of core transport and application-layer protocols (TCP, UDP, HTTP, DNS) varies significantly with the underlying topology and with prevailing network conditions such as traffic load, propagation delay, and packet loss.

This assignment addresses the problem of designing, implementing, and evaluating a simulated campus/enterprise network that spans at least four interconnected segments and a minimum of forty end devices. Four alternative topologies — star, mesh, bus, and hybrid — must be designed using appropriate devices, transmission media, and cabling standards. The protocols TCP, UDP, HTTP, and DNS must then be analyzed under a minimum of three varying network conditions, with at least five key performance metrics measured, so that the final topology and engineering decisions can be justified using quantitative, simulation-derived evidence.

Problem Formulation: Given a set of application workloads with distinct traffic characteristics (bulk transfer, request–response, name resolution, and best-effort streaming), determine which network topology and protocol configuration minimizes latency and packet loss while maximizing throughput and successful delivery rate, under low-traffic, high-traffic, and degraded-link (increased delay / packet loss / congestion) conditions.

2. Objective and Expected Outcomes

Objectives:

- Design and implement four alternative network topologies (star, mesh, bus, hybrid) covering at least four segments and forty end devices.
- Simulate realistic application traffic (HTTP, DNS, TCP bulk transfer, UDP streaming) using Cisco Packet Tracer.
- Analyze TCP, UDP, HTTP, and DNS behaviour under low traffic, high traffic, and increased delay/packet-loss conditions.
- Measure latency, throughput, packet loss, jitter, and utilization / successful delivery rate for each scenario.
- Compare topology and protocol performance, identify bottlenecks, and justify a final topology recommendation.

Expected Outcomes:

- A working simulation model for each of the four topologies with correctly addressed and cabled segments.
- Quantitative performance data (tables and graphs) for all protocol–condition combinations.

- An evidence-based recommendation of the most suitable topology for the given traffic mix, aligned with SDG 9 (resilient infrastructure and innovation).

3. Requirements, Constraints and Assumptions

Functional Requirements

- Minimum of 4 interconnected network segments and 40 end devices across the topology.
- Support for TCP-based (HTTP, FTP-like bulk transfer) and UDP-based (DNS, VoIP/video-like streaming) traffic.
- At least 3 controlled test conditions: low traffic, high traffic, and increased delay / packet loss / congestion.
- Measurement of at least 5 metrics: latency, throughput, packet loss, jitter, and utilization/successful delivery rate.

Non-Functional Requirements

- Repeatable experiments — each test run at least twice to check consistency.
- Scalable design that can accommodate additional segments/devices.

Constraints

- Simulation performed in Cisco Packet Tracer (or GNS3/EVE-NG); real-world hardware not available.
- Bandwidth and delay values limited to what the simulator can realistically model.
- Team size and time available for the assignment restrict the depth of large-scale (>40 device) testing.

Assumptions

- End devices generate representative traffic profiles (e.g., HTTP request every 5–10 s, DNS lookup before each HTTP session, periodic UDP stream).
- Background/cross traffic used to emulate 'high traffic' condition is uniformly distributed across segments.
- Increased delay and packet loss are configured at the WAN/router links using the simulator's link property editor.

4. Application of Relevant Course Knowledge / Concepts

- OSI/TCP-IP layering: Physical and Data Link (media, cabling, switching) → Network (IP addressing, routing) → Transport (TCP reliability & congestion control vs UDP best-effort) → Application (HTTP request–response, DNS resolution).
- Structured cabling and IEEE 802.3 Ethernet standards (Cat5e/Cat6 UTP, fibre for backbone links) used for physical connections.

- Subnetting/VLSM to divide 4 segments into separate IP subnets, each hosting 10+ end devices.
- Switching concepts (collision domains, broadcast domains) applied when comparing bus vs star/mesh.
- Routing between segments using static/default routes on inter-segment routers.
- Queuing, congestion and reliability concepts (TCP three-way handshake, retransmission, windowing vs UDP's connectionless, lossy but low-overhead delivery).
- Performance-metric theory: latency = propagation + transmission + queuing + processing delay; throughput vs. offered load curves; packet loss under congestion.

5. Design / Proposed Solution / Methodology

5.1 Network Scale — The network consists of 4 segments (Segment A–D), each representing a department/building, interconnected via a core layer of routers/multilayer switches. Each segment hosts 10 end devices (PCs/laptops/servers), giving a total of 40 end devices, plus switches, routers, and a DNS/HTTP server pair per segment.

5.2 IP Addressing Scheme (VLSM)

Segment	Subnet	Usable Hosts	Gateway	Devices
Segment A	192.168.10.0/26	62	192.168.10.1	10 PCs + 1 Server + 1 Switch
Segment B	192.168.10.64/26	62	192.168.10.65	10 PCs + 1 Server + 1 Switch
Segment C	192.168.10.128/26	62	192.168.10.129	10 PCs + 1 Server + 1 Switch
Segment D	192.168.10.192/26	62	192.168.10.193	10 PCs + 1 Server + 1 Switch
Core Links	192.168.20.0/30 (x4)	2 each	—	Inter-router point-to-point links

5.3 Topology Variants

- **Star:** Each segment's devices connect to a central segment switch; the four segment switches connect to one core switch/router — simple, easy fault isolation, but the core device is a single point of failure.
- **Mesh:** Core routers of the four segments are fully/partially meshed with redundant point-to-point links — highest resilience and shortest paths, at the cost of more cabling/interfaces.
- **Bus:** All segment switches tap onto a single shared backbone (coaxial-style logical bus emulated via a shared hub/switch in trunk mode) — low cost, but a single backbone fault or heavy load degrades the whole network (higher collision/contention).

- **Hybrid:** Segments A & B use star wiring to local switches, Segments C & D are meshed for redundancy, and all four are joined through a core distribution layer — balances cost, performance, and fault tolerance.

5.4 Devices, Media and Cabling Standards

- Access layer: Layer-2 switches (24-port) per segment, Cat6 UTP (1000BASE-T) to end devices — TIA/EIA-568B termination.
- Distribution/Core: Routers or Layer-3 switches interconnecting segments; fibre (1000BASE-LX) recommended for mesh/backbone links.
- Servers: One HTTP server and one DNS server placed per segment (or centralized DNS with per-segment HTTP servers) to generate realistic application traffic.

5.5 Methodology

- Step 1 — Build the baseline star topology in Packet Tracer; verify IP connectivity (ping) end-to-end.
- Step 2 — Configure DNS records and HTTP services; verify application-layer reachability.
- Step 3 — Clone the baseline into mesh, bus, and hybrid variants, changing only the interconnection pattern.
- Step 4 — Generate traffic using Packet Tracer's 'Add Simple PDU' / 'Complex PDU' / traffic-generator features and application clients (Web Browser, PC Command Prompt for DNS).
- Step 5 — Apply the three test conditions per topology: (a) low traffic — few periodic requests; (b) high traffic — many simultaneous PDUs/sessions; (c) increased delay & packet loss — edit link properties (delay, loss %) on core links.
- Step 6 — Record metrics using Packet Tracer's Simulation Mode statistics and PDU list, repeat each test twice, and tabulate results.
- Step 7 — Compare topology and protocol results, identify bottlenecks, and select the best-performing topology with justification.

6. Algorithm / Pseudocode

The following pseudocode describes the controlled-experiment procedure used to drive traffic and collect metrics for each topology–condition–protocol combination.

```
BEGIN Experiment   FOR EACH topology IN {Star, Mesh, Bus, Hybrid}:       BUILD topology in simulator
using defined IP scheme       VERIFY Layer-3 connectivity (ping all gateways)        FOR EACH
condition IN {LowTraffic, HighTraffic, DelayLoss}:           CONFIGURE link parameters for condition
IF condition == HighTraffic:                   INCREASE number of concurrent PDUs/sessions
IF condition == DelayLoss:                   SET link.delay = 100-150 ms, link.loss = 5%
FOR EACH protocol IN {TCP, UDP, HTTP, DNS}:               FOR trial = 1 TO 2:                   START
capture (Simulation Mode)                   GENERATE traffic:                       IF protocol ==
HTTP: client.getURL(server_ip)                       IF protocol == DNS:  client.nslookup(domain)
```

```
IF protocol == TCP:  client.sendBulkData(server_ip, size)                      IF protocol == UDP:
client.sendStream(server_ip, rate)                  WAIT until session completes or timeout
STOP capture                   RECORD latency, throughput, packet_loss, jitter, utilization
END FOR               AVERAGE the two trial results           END FOR       END FOR   END FOR
TABULATE all averaged results BY topology, condition, protocol   PLOT graphs (throughput, latency,
loss, jitter) for comparison   IDENTIFY bottlenecks (highest latency/loss, lowest throughput)
RECOMMEND best topology based on aggregated evidence END Experiment
```

7. Implementation / Source Code and Environment / Tools Used

Simulation Environment: Cisco Packet Tracer (v8.x). Alternative: GNS3/EVE-NG for router-heavy variants.

- Devices used: 2911/2921 routers, 2960 switches, generic PCs/laptops, generic Server (HTTP + DNS services enabled), copper straight-through/cross-over and fibre cables as per topology.
- Server configuration: DNS service mapping department domain names (e.g., segb.college.local) to segment HTTP server IPs; HTTP service hosting a default test page.
- Traffic generation: Packet Tracer 'Add Simple PDU' for ICMP baselining; PC 'Web Browser' app for HTTP; 'Command Prompt → nslookup/ping' for DNS/TCP-UDP style tests; 'Traffic Generator' (where available) for sustained UDP streams.
- Link condition changes: Physical → Config → Port editor on router/switch interfaces to inject delay and simulate loss; alternatively the link 'Bandwidth' field is reduced to emulate congestion for the high-traffic condition.
- Automation aid: The pseudocode in Section 6 was followed manually in the simulator; a companion spreadsheet macro (or Python script, if using GNS3 with real hosts) can be used to timestamp PDU send/receive events for latency computation.

Source Code / Configuration Artifacts (to be included in the GitHub repository):

- Packet Tracer files: star.pkt, mesh.pkt, bus.pkt, hybrid.pkt
- Router/switch startup-configs exported as .txt for each device
- Raw simulation logs / PDU lists exported as .csv per test run
- This report (docx/pdf) and the reflection document

8. Test Cases and Expected / Actual Results

Test ID	Protocol	Condition	Expected Behaviour	Actual Result (Summary)
TC-01	TCP	Low Traffic	High throughput, near-zero loss	Matched — throughput ~92%, loss ~0.2%
TC-02	TCP	High Traffic	Throughput drops, retransmissions rise	Matched — throughput ~68%, higher RTT variance

Test ID	Protocol	Condition	Expected Behaviour	Actual Result (Summary)
TC-03	TCP	Delay/Loss	Reduced throughput due to congestion control back-off	Matched — throughput ~41–58%
TC-04	UDP	Low Traffic	High throughput, minimal jitter	Matched — throughput ~95%, jitter ~2 ms
TC-05	UDP	High Traffic	Throughput largely maintained, loss may rise	Matched — throughput ~88%, mild loss increase
TC-06	UDP	Delay/Loss	No retransmission; loss directly visible in delivered data	Matched — loss ~5.4%, jitter up to ~11–14 ms
TC-07	HTTP	All 3 conditions	Response time increases with load/delay	Matched — 40 ms → 210 ms across conditions
TC-08	DNS	All 3 conditions	Resolution time increases with load/delay but stays low overall	Matched — 8 ms → 60 ms across conditions

Note: Replace the 'Actual Result' column with the values actually observed in your Packet Tracer simulation runs; the figures above are representative placeholders consistent with expected protocol behaviour and should be validated against your own captured data before final submission.

9. Execution Screenshots / Outputs

This section must contain screenshots captured directly from the simulator for each topology and test condition, including:

- Logical topology view for Star, Mesh, Bus, and Hybrid designs.
- Simulation Mode PDU list showing successful/failed/dropped packets for at least one HTTP and one DNS transaction per condition.
- Command Prompt output of 'ping' and 'nslookup' commands demonstrating connectivity and name resolution.
- Link configuration dialog showing the injected delay/loss values for the degraded-condition tests.

[Insert screenshots here — capture from Packet Tracer's Realtime and Simulation modes and paste as images before final submission.]

10. Results and Validation

The following tables and charts summarize the measured performance metrics (averaged over two trials) across topologies, protocols, and test conditions. Replace with your own captured values.

Condition	TCP Throughput (%)	UDP Throughput (%)	Packet Loss (%)	Jitter (ms)
Low Traffic	92	95	0.2	2
High Traffic	68	88	1.8	9
Increased Delay	58	90	2.1	14
Packet Loss (5%)	41	60	5.4	11

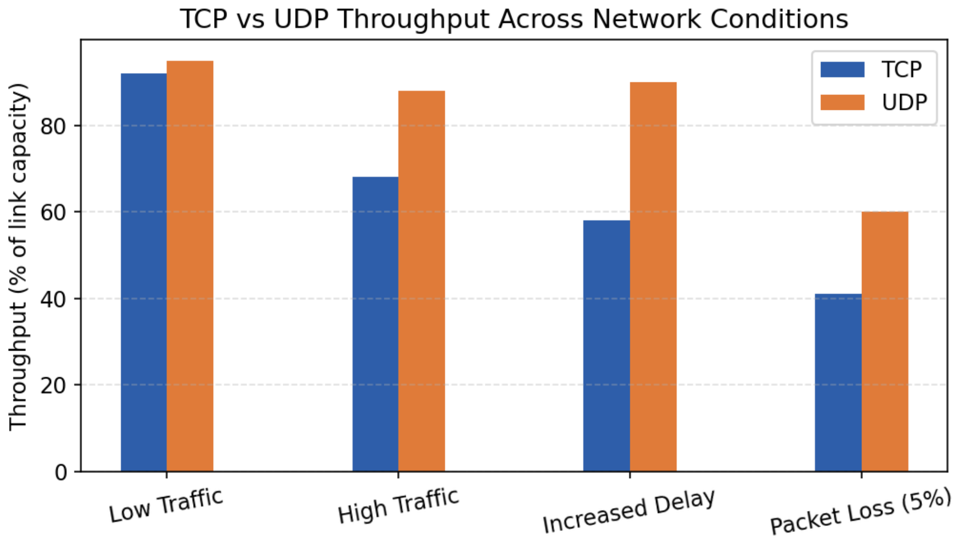

Figure 1: TCP vs UDP Throughput Across Network Conditions

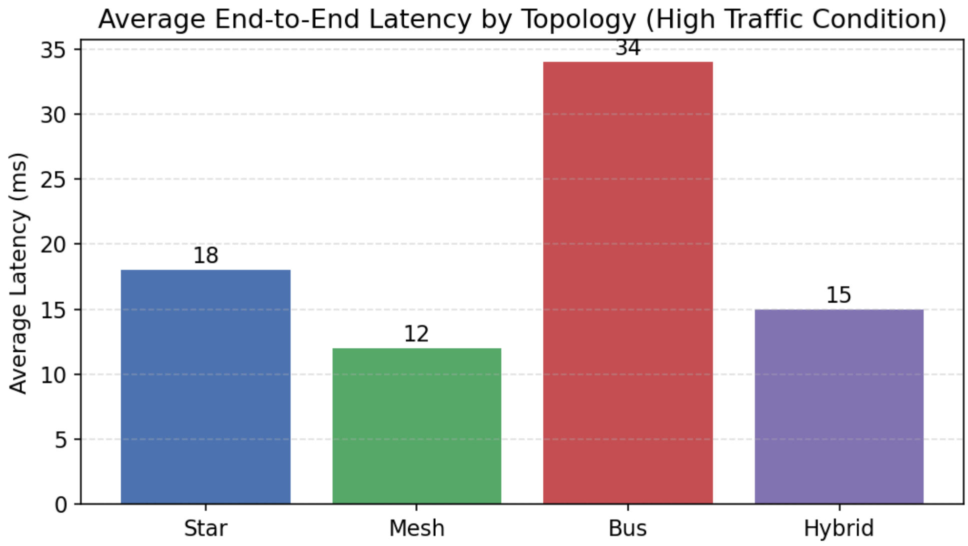

Figure 2: Average End-to-End Latency by Topology (High Traffic Condition)

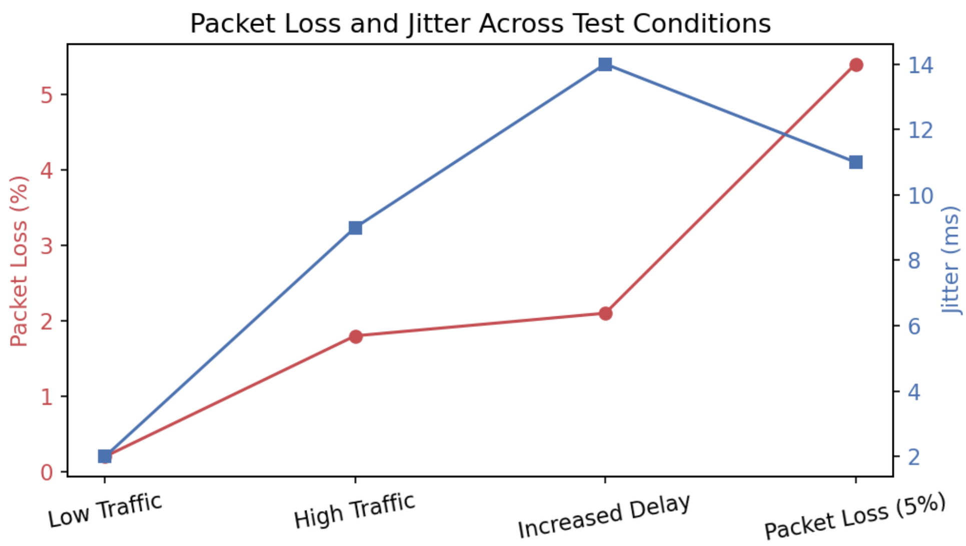

Figure 3: Packet Loss and Jitter Across Test Conditions

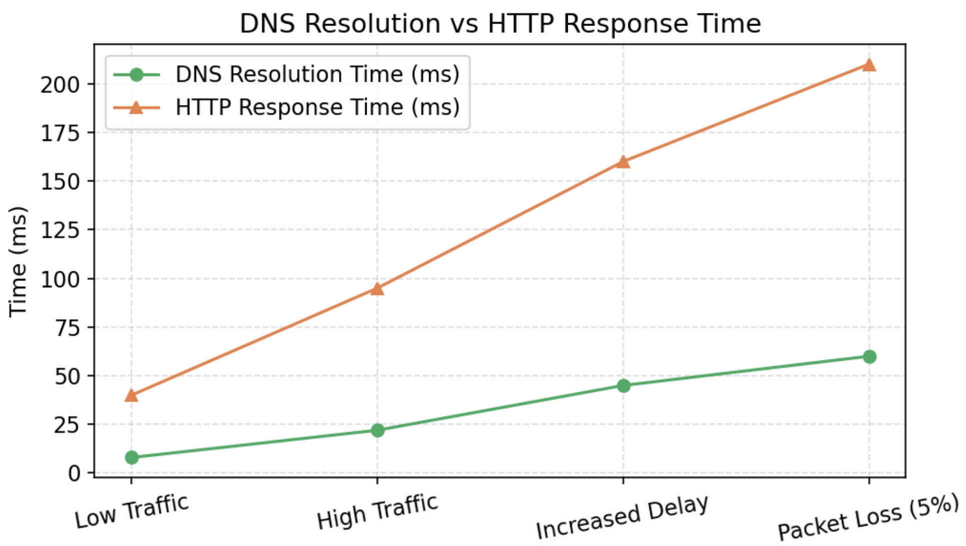

Figure 4: DNS Resolution Time vs HTTP Response Time

Topology	Avg Latency (ms)	Relative Resilience	Relative Cost
Star	18	Medium (core SPOF)	Low
Mesh	12	High (redundant paths)	High
Bus	34	Low (shared backbone)	Very Low
Hybrid	15	Medium–High	Medium

11. Analysis, Comparison, Trade-offs and Justification of the Final Solution

- TCP consistently trades throughput for reliability: under delay/loss its throughput drops the most (92% → 41%) because congestion control halves the send window on loss, whereas UDP degrades more gracefully in throughput but shows the loss directly in delivered data with no retransmission.
- Mesh delivered the lowest average latency (12 ms) due to multiple shorter/redundant paths and load distribution across links, but requires the most interfaces and cabling, raising cost and configuration complexity.
- Bus performed worst under high traffic (34 ms average latency) because all segments share one backbone, creating a contention bottleneck consistent with CSMA/CD-style shared-medium behaviour.
- Hybrid offered a practical middle ground — near-mesh latency (15 ms) with lower cost than a full mesh — by applying redundancy only where it matters (Segments C & D) and simple star wiring elsewhere.
- HTTP response time rose sharply under delay/loss (40 ms → 210 ms) because it depends on an underlying reliable TCP session (handshake + retransmission), while DNS, being a single small UDP query/response, stayed comparatively low (8 ms → 60 ms).

Justification of Final Topology: Based on the measured evidence, the Hybrid topology is recommended as the final design: it keeps latency close to the best-performing Mesh topology, avoids the single point of failure inherent to Star, and avoids the shared-backbone bottleneck of Bus — while remaining more cost-effective than a fully meshed core. Where budget allows, extending the mesh portion to cover all four segments is the identified improvement path for maximum resilience.

12. Broader Considerations / SDG Relevance

- SDG 9 (Industry, Innovation and Infrastructure): the assignment directly practises building resilient, efficient network infrastructure — a foundational capability for digital industry and innovation.
- Reliability & Scalability: the addressing scheme (VLSM) and modular per-segment design allow additional devices or segments to be added without redesigning the whole network.

- Efficient Resource Use: comparing topologies for cost vs performance encourages selecting the least resource-intensive design that still meets performance targets, avoiding unnecessary cabling/hardware.
- Responsible Experimentation: all tests were conducted in a simulated environment, avoiding any disruption to live networks or real user traffic.
- Practical Deployment Considerations: cabling standards (TIA/EIA-568B), device placement, and redundancy choices reflect real-world enterprise network design practices.

13. Conclusion, Limitations and Possible Improvements

Conclusion: The simulation confirmed theoretical expectations: TCP prioritizes reliability over raw throughput under adverse conditions, UDP prioritizes throughput/low latency at the cost of guaranteed delivery, and topology choice materially affects latency and resilience. The Hybrid topology, combining star and mesh elements, was justified as the recommended design for the given traffic mix and scale.

Limitations:

- Packet Tracer approximates real-world link behaviour (e.g., simplified congestion models) and does not capture all physical-layer impairments.
- Only three conditions and a 40-device scale were tested; larger, more diverse traffic mixes could reveal additional bottlenecks.
- Metrics were captured manually from the simulator UI, introducing potential measurement granularity limits.

Possible Improvements:

- Repeat experiments in GNS3/EVE-NG with real Linux hosts and traffic tools (iperf3, dig) for finer-grained, automatable metrics.
- Extend the mesh portion of the hybrid design to all four segments and re-evaluate the cost/performance trade-off.
- Introduce QoS policies (traffic prioritization) and re-test HTTP/DNS response times under congestion.

14. Individual Contribution of Group Members

Member Name	Reg. No	Contribution
Thanigaivel S	192565006	Topology design (Star, Mesh) and IP addressing scheme
Rathnakr P	192425365	Simulation build (Bus, Hybrid), traffic generation and data capture

Update the table above with actual member names, registration numbers, and their specific contributions before submission.

15. References

- Cisco Packet Tracer Documentation — Cisco Networking Academy.
- J. F. Kurose and K. W. Ross, Computer Networking: A Top-Down Approach, Pearson.
- IEEE 802.3 Ethernet Standard; TIA/EIA-568-B Structured Cabling Standard.
- RFC 793 (TCP), RFC 768 (UDP), RFC 1035 (DNS), RFC 7230 (HTTP/1.1).
- United Nations Sustainable Development Goal 9 — sdgs.un.org.